

App Dev Project Report

Placement Portal Application

1.1 Student Details

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About Me: I am a student in the IIT Madras BS Degree program with a strong interest in full-stack development and building real-world applications.

1.2 Project Details

Project Title: HireTrack: Placement Portal Application

Problem Statement:

Traditional placement systems rely on manual processes, spreadsheets, and emails, which leads to inefficiency, poor communication, and lack of transparency.

Students cannot easily track applications, companies face difficulty managing applicants, and admins struggle with approvals and monitoring.

It face several challenges like:

- **Inefficient Management:** Paper-based or disconnected systems make it difficult to manage placement drives, applications, and student information
- **Poor Communication:** Students struggle to find relevant job opportunities and track their application status
- **Administrative Burden:** HR personnel and admin staff spend excessive time managing drives, approving companies, and processing applications manually
- **Lack of Transparency:** No centralized platform for students to view drives, apply, and track their progress through the hiring process
- **Scalability Issues:** Manual systems cannot handle multiple concurrent placement drives efficiently

Approach:

To solve the problem, a role-based web application was designed for Admin, Company, and Student users.

First, the requirements of all three users were analyzed, such as managing placement drives, tracking applications, and approving companies. Then, a database structure was created with entities like Student, Company, Placement Drive, and Application.

The system was developed using Flask for backend and Jinja2 with Bootstrap for frontend. Authentication and role-based access control were implemented to ensure secure access.

Core features like company approval, placement drive creation, student applications, and status tracking were implemented. Constraints such as preventing duplicate applications and allowing only approved companies to create drives were also enforced.

Finally, the system was tested to ensure smooth functionality and proper handling of different user workflows.

1.3 AI/LLM Declaration

I have used ChatGPT in a limited capacity during the development of this project. The tools were primarily used for debugging errors, understanding Flask concepts, structuring routes, and improving documentation.

AI assistance was also used for minor code suggestions and UI improvements using Bootstrap.

The approximate usage of AI/LLM tools is around 15–20%. The remaining work, including application logic, database design, feature implementation, and overall system development, has been completed independently by me.

I confirm that this project is my original work and AI tools were used only as supportive aids.

1.4 Technologies Used

The following technologies and tools were used in the development of this project:

- **Flask:** Used as the backend framework to handle server-side logic and routing
- **Flask-SQLAlchemy:** Used for database modeling and interaction
- **Flask-Login:** Implemented for authentication and session management
- **SQLite:** Used as the database for storing application data
- **Jinja2:** Used for rendering dynamic HTML templates
- **HTML, CSS, Bootstrap:** Used for designing a responsive and user-friendly interface
- **Werkzeug Security:** Used for password hashing and security

1.5 Database Schema

The database for the Placement Portal was designed to manage students, companies, placement drives, and applications efficiently. It follows a relational structure with proper use of primary and foreign keys to maintain data integrity.

The main entities in the system are:

- Student
- Company
- Admin
- Placement Drive
- Application

Each entity stores relevant information such as student details, company profiles, job descriptions, and application status.

Relationships:

- A **Student** can apply to multiple placement drives
- A **Company** can create multiple placement drives
- A **Placement Drive** can have multiple student applications
- The **Application** table acts as a link between Student and Placement Drive

Constraints:

- One student can apply to a drive only once
- Only approved companies can create placement drives
- Proper foreign key relationships ensure consistency

1.6 Application Features

The application includes the following key features:

- Role-Based Access Control (RBAC): Separate access and functionalities for Admin, Company, and Student users.
- Authentication System: Secure login and registration system with password hashing.
- Placement Drive Management: Companies can create, edit, and manage placement drives after admin approval.
- Application System: Students can apply for placement drives and track their application status.
- Admin Control Panel: Admin can approve/reject companies and placement drives, and manage all users.
- Duplicate Application Prevention: Ensures that a student can apply to a specific drive only once.
- Resume Management: Students can upload resumes and companies can download them.
- Status Tracking: Application status updates (Applied, Shortlisted, Selected, Rejected) are visible to students.
- Responsive User Interface: Built using Bootstrap to ensure usability across devices.

1.7 Architecture and Features

System Architecture:

The application follows a three-tier architecture, consisting of:

- Presentation Layer (Frontend): Built using HTML, CSS, Bootstrap, and Jinja2 templates to provide an interactive user interface.
- Application Layer (Backend): Developed using Flask, which handles routing, business logic, authentication, and role-based access control.
- Database Layer: SQLite database is used to store and manage data related to students, companies, placement drives, and applications.

The system follows a role-based workflow, where Admin, Company, and Student interact with the application through their respective dashboards.

1.8 Video Presentation

Drive Link:

<https://drive.google.com/file/d/1FN4zxCuZF83vzyFNgm02oJ-YyEyJ8Se8/view?usp=sharing>